DAIFEND

SECURING THE AI RUNTIME

AI SECURITY INFRASTRUCTURE

PLATFORM BRIEF — ENTERPRISE AI-NATIVE DEFENSE

The operating system for AI security.

Traditional cybersecurity protects infrastructure. **Daifend protects AI systems**: AI agents, vector databases, RAG pipelines, autonomous reasoning chains, and long-term memory.

AI MEMORY TRUST

0–100 TRUST SCORE

Quantifies integrity of persisted embeddings: drift, poisoning signals, provenance, and replay safety.

AGENT RUNTIME POSTURE

Live CONTAINED

Tracks autonomy risk: permission boundaries, unsafe tool attempts, and containment effectiveness.

THREAT TELEMETRY

Signatures AI-NATIVE

Prompt injection, semantic corruption, agent tool misuse, model manipulation, synthetic identity.

Positioning

Daifend is not a SIEM. It is a **security control plane** for AI runtime integrity: detection + policy enforcement + autonomous repair, tied together by high-fidelity telemetry.

Core modules and how they work

1. AI MEMORY INTEGRITY CENTER

Protects vector databases and long-term memory from manipulation, poisoning, and semantic drift.

- **Embedding trust scoring** per namespace with baselines, entropy checks, and signature matches.
- **Semantic drift analysis** across time windows to detect subtle meaning shifts.
- **Suspicious injection detection** for anomalous write patterns and lineage breaks.
- **Memory lineage explorer** from ingestion → chunking → embedding → retrieval.
- **Rollback** to verified baselines and quarantine suspect segments.

2. AI AGENT SECURITY RUNTIME

Observes autonomous agent sessions and enforces policies around tools, permissions, and reasoning.

- **Runtime containment**: isolate sessions, rotate credentials, block unsafe execution attempts.
- **Policy enforcement**: risk-based tool permissions, escalation gates, and audit trails.
- **Reasoning-chain auditability**: correlate model outputs to tool calls and memory reads/writes.
- **Behavior fingerprinting**: detect session drift, abnormal communication patterns, or stealth pivots.

3. RAG & VECTOR SECURITY

Secures ingestion and retrieval paths where prompt injection and malicious documents enter.

- **Prompt injection detection** with instruction-hierarchy anomaly checks.
- **Malicious document scanning** (payload patterns, policy violations, disallowed instructions).
- **Retrieval chain validation** with signed context windows and provenance enforcement.
- **Poisoned embedding detection** via trust score deltas and semantic corruption signatures.

4. AI THREAT INTELLIGENCE

AI-native signature feed to detect and explain modern threats unique to AI systems.

- **Attack signatures** (prompt-borne malware, retriever bypass, spec inversion, identity spoofing).
- **Evolution tracking**: signature families, mutation paths, recommended mitigations.
- **Model manipulation detection** signals tied to runtime policy and incident correlation.

5. SELF-HEALING ENGINE

Autonomous repair workflows that isolate the blast radius and restore trusted state.

- **Isolation**: quarantine vector segments, gate retrieval, rotate agent sessions.
- **Rollback orchestration** across memory snapshots and runtime sessions.
- **Trust restoration** with validation gates and auditable change sets.
- **Incident playback**: end-to-end narrative across detection → response → recovery.

6. COGNITIVE FIREWALL

Detects persuasion and emotional manipulation vectors in AI-mediated conversations.

- **Persuasion analysis**: pressure, urgency, authority cues, and psychological steering.
- **Emotional intent**: identify manipulative sentiment trajectories.
- **Deepfake conversation analysis**: signal inconsistencies and synthetic identity patterns.

Live AI Attack Simulation (investor-demo flow)

SCENARIO

A malicious document enters ingestion, contaminates vector memory, and manipulates an agent via retrieved context. Daifend detects drift + signatures, isolates corruption, heals, and generates an incident report.

PROMPT INJECTION

MEMORY POISONING

DRIFT CORRELATION

AUTONOMOUS ROLLBACK

OPERATIONAL OUTPUTS

- **Trust score timeline** and semantic drift heatmap.
- **Threat ticker** with AI-native signatures and surfaces.
- **Healing actions feed** (isolate, rotate, rollback, invalidate, restore baseline).
- **Structured incident report** with evidence, actions, and recommended controls.

STEP-BY-STEP

#	Event	Daifend detection	Response
1	Malicious document uploaded	Pre-ingestion risk scan flags hierarchy anomalies	Gate ingestion, begin deep scan
2	Vector DB becomes poisoned	Trust score drops, drift escalates, lineage delta spike	Quarantine suspect segment
3	Agent retrieves manipulated memory	Unsafe tool attempt signatures appear	Contain session, rotate agent identity
4	Anomaly confirmed	Signature + drift correlation crosses threshold	Escalate to incident
5	Runtime isolation	Control plane enforces retrieval chain constraints	Freeze writes, block risky tools
6	Self-healing restores integrity	Validation gates pass against trusted baseline	Rollback + invalidate embeddings
7	Incident report generated	Evidence assembled with provenance	Export to SOC / ticketing

Architecture: where Daifend sits

CONTROL PLANE

Daifend runs alongside AI infrastructure as a policy + telemetry + response layer.

- **Agents**: permissions, tool calls, reasoning traces, containment.
- **RAG**: ingestion scanning, retriever validation, context provenance.
- **Vector memory**: trust scoring, lineage, drift, rollback.
- **Threat intel**: signatures and evolving attack families.

INTEGRATIONS (MOCK)

- OPENAI · AZURE OPENAI · BEDROCK · ANTHROPIC
- PINECONE · WEAVIATE · LANGCHAIN
- KUBERNETES · CROWDSTRIKE · MICROSOFT SENTINEL

Outputs: incidents, alerts, reports, and integrity metrics for SOC visibility without becoming a generic SIEM UI.